



OPERATOR GUIDE

Box3D Metal Quick Start

Reconstruct the pinned Box3D checkout, apply the Metal overlay, validate it, and enable GPU routing per world.



Reconstruct and run

The public repository is an overlay: it contains no Box3D checkout. The bootstrap script clones the canonical upstream revision, verifies the patch, applies it, and builds a Release configuration.

Requirements

- Apple Silicon Mac and macOS Metal
- Xcode command-line tools
- Git, CMake, and Ninja

```
git clone https://github.com/Lulzx/box3d-metal.git
cd box3d-metal
./scripts/bootstrap.sh ../box3d-metal-worktree
```

Validation

```
../box3d-metal-worktree/build/metal-release/bin/test MetalTest
../box3d-metal-worktree/build/metal-release/bin/metal_demo
```

For a disposable end-to-end reconstruction, run `./scripts/verify-clean.sh`.

Enable per world

```
b3WorldId world = b3CreateWorld(&worldDef);
if (!b3World_EnableMetal(world, 32768)) {
/* CPU path remains usable. */
}
```

The body threshold is deliberately caller-selected. It is not a universal recommendation; measure your world with full-step timing.

Experimental residency stages

```
b3World_SetMetalFinalization(world, true);
b3World_SetMetalBroadPhase(world, true);
```

Finalization computes body and awake-shape results on Metal. Broad-phase mode updates resident leaves, refits internal bounds, and compacts exact-order candidates; CPU topology, filtering, and contact creation remain.

Read route telemetry

```
b3MetalProfile p = b3World_GetMetalProfile(world);
printf("%s contacts=%llu pairs=%llu pairFallbacks=%llu\n",
p.deviceName, p.contactDispatchCount,
p.pairDispatchCount, p.pairFallbackCount);
```

Dispatch counters prove a Metal stage ran. Pair dispatch means tree traversal and compaction, not GPU tree ownership, filtering, narrow phase, CCD, sleeping, or events.

Next references

- docs/compatibility.md - exact supported and CPU fallback surface
- docs/performance.md - measured M4 Pro crossovers
- docs/troubleshooting.md - initialization, routing, and performance diagnosis